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MILITARY SPECIFICATION

DE-ICING SYSTEM, PNEUMATIC BOOT, AIRCRAFT, GENERAL SPECIFICATION FOR

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the general requirements for pneumatic de-icing systems for wings, empennages, radomes, radio masts, air induction system entrance cones, ducts, etc., of aircraft. De-icing boots shall be of one type, operating at 15- to 22-psig pressure.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585 — Boxes; Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner

PPP-B-636 — Boxes, Fiber.

MILITARY

JAN-P-100 — Packaging and Packing for Overseas Shipment — General Specification.

MIL-C-104 — Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.

MIL-P-116 — Preservation, Methods of.

MIL-B-138 — Boxes, Wood, Fiberboard-Lined for Overseas Shipment (for Weight of Contents not Exceeding 500 pounds).

MIL-D-5028 — Drawing and Data Lists: Preparation of Manufacturers' (for Production Aeronautical and Associated Equipment).

MIL-E-5272 — Environmental Testing, Aeronautical and Associated Equipment, General Specification for.

MIL-E-7894 — Electric Power, Aircraft, Characteristics of.

MIL-P-7936 — Parts and equipment, aeronautical, preparation for delivery.

FSC 1650

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MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

STANDARDS

MILITARY

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U. S. Military Property.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to beginning quantity production, preproduction samples shall be subjected to preproduction testing. (See 4.2.1 and 6.2).

3.2 Components. The de-icing boot system shall consist of the following components: De-icing boots, air and vacuum source, valves, oil separator (if required), timer, test connection, and gages.

3.3 Materials. Materials shall conform with the applicable specifications and shall be as specified herein. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest permissible weight, and suitable for the intended purpose. The manufacturer shall furnish a specification for approval by the procuring agency covering the material listed in paragraph 3.14.2.

3.4 Maintenance. All parts which require periodic inspection or replacement shall be accessible for inspection and readily removable to facilitate maintenance. Pipe threads shall not be used for any connection that must be disconnected to permit removal or maintenance of any component of the system.

3.5 Environment. All components used in the system shall be capable of operating satisfactorily under the extremes of temperature and altitude for which the aircraft is designed and shall not be deleteriously affected by exposure to salt spray, humidity, vibration (not applicable to boots), fungus, and sunshine within exposure time limits specified by manufacturing and approved by procuring agency (for boots only). Any equipment installed in locations where explosive vapors may exist shall be safe for operation in this location.

3.6 Design and construction.

3.6.1 *De-icing boots.* The de-icing boots shall be designed with pneumatically inflated tubes which, when fully expanded, do not exceed $1\frac{1}{4}$ inches in diameter. Inflatable tubes shall operate at a pressure of 15 to 22 psig and shall extend chordwise along the leading edges of the wings, fins, stabilizers, radio masts, radomes, scoops, etc. If thickness of air foil section is insufficient for proper performance of de-icing boots with inflatable tubes extending chordwise the boots shall be provided with inflatable tubes extending in the spanwise direction.

3.6.2 *Air and vacuum source.* An air and vacuum source shall be provided which will meet the pressure and vacuum requirements of the de-icing boot on the specific aircraft. When a separate air pump is used, it shall operate the boot de-icing system at the design pressure from sea level to 25,000 feet. In turbopropeller or turbojet aircraft, the engine compressor air may be used as the pressure source and air from the same pressure source used in an ejector to provide the required vacuum. A heat exchanger (preferably consisting of the pressure manifold itself) shall be provided when required to prevent allowable boot inlet temperature being exceeded.

3.6.3 *Electrical power.* Electrical power required in the operation of the de-icing system shall be in accordance with Specification MIL-E-7894.

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3.6.4 *Valve (safety).* Safety valves, or valve, shall be installed to prevent damage to the system in the event of failure of the pressure-regulating valve. The safety valves shall be set to relieve at least 3 psi, but no greater than 5 psi above the normal operating pressure of the system.

3.6.5 *Oil separator (primary and secondary).* The primary oil separator shall remove 75 percent of the oil from the air in the deicer system and shall return the oil to the engine sump. The secondary oil separator shall remove the remaining oil from the system. The requirement for an oil separator in a turbocompressor bleed system shall be individually determined.

3.6.6 *Valve (check).* On multiengine aircraft, check valves shall be installed in the appropriate manifolds or compressor bleed lines to insure the operation of the system in the event of an engine failure.

3.6.7 *Valve (regulating and unloading).* The predetermined pressure shall be regulated to vary not more than ± 0.5 psi. The capacity of the unloading valve shall be sufficient to dump overboard the output of the pump when the engine is operating at takeoff rpm. The pressure-regulating valve in a turbocompressor bleed-type system shall not incorporate dump provisions.

3.6.8 *Valve (distribution).* The distributor valve, or valves in the case of a manifold-type system, shall operate instantaneously with a minimum of pressure drop.

3.6.9 *Timer.* The timer shall operate the system automatically to meet the requirements of 3.11 except that the time required to inflate and deflate the pneumatic tubes shall be as rapid as possible in order to minimize the effects on the performance of high-speed aircraft.

3.6.10 *Valve (vacuum relief).* On a system which uses the intake side of the air pump as the vacuum source, a vacuum relief valve shall be provided with sufficient capacity to regulate the vacuum pressure to ± 0.5 inches Hg of the design value when the engine is operating at takeoff rpm. When the vacuum is provided by an ejector, a vacuum relief valve is not required.

3.6.11 *Test connection.* A test connection, or connections, shall be provided in the system for ground checking.

3.6.12 *Gages.* A pressure and vacuum gage shall be provided in multiplace aircraft. In single-place aircraft the requirement for a pressure and vacuum gage shall be individually determined. When these gages are mounted in a pressurized compartment, the gages shall measure the difference between system and pressure ambient to airplane.

3.7 Icing conditions. The de-icing boot system shall be capable of removing, from all protected surfaces of the aircraft, ice accretions which form under conditions at least equivalent in severity to those in which the free air temperature is 15° F, or higher, the liquid water content of the air is 0.5 gram per cubic meter of air, with the average diameter of water droplets of 20 microns. Chordwise coverage shall be adequate for the effective removal of ice accretions under all flight conditions including flight speeds from minimum cruise or loiter, to maximum speed of the aircraft.

3.8 Airfoil coverage. De-icing boot coverage shall extend from fuselage to nacelle, from nacelle to nacelle, from nacelle to around the tip of the wing or stabilizer surface, or from fuselage to and around engine duct, and from engine duct to the tip of the wing. The entire empennage surface-leading edge shall be protected by de-icing boots. Chordwise coverage shall amount to a maximum of 12 percent and a minimum of 7½ percent except in locations where it is impracticable because of leading edge cutouts or other interfering structure, unless the airframe contractor can substantiate by a droplet trajectory analysis on the particular airfoil that more or less coverage is required.

3.9 Attachment. De-icing boots shall be attached to the surfaces of the aircraft to be

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protected by cementing, or other method acceptable to the procuring activity. The cementing materials used shall permit the removal and reuse of the de-icing boots without damage either to the de-icing boots themselves or to the aircraft.

3.10 **Boot thickness.** The overall average thickness of the de-icing boot shall be 0.125 ± 0.010 inch, except in the manifold areas, where the maximum boot thickness may be .156 ± .015. De-icing boots shall be designed to withstand, for 5 minutes maximum without distortion, blowout, or any other damage, an internally applied pressure of 5 psi above the normal design pressure.

3.11 **Controls.** Controls shall be provided for the de-icing boot system which will provide for a symmetrical operation of the boots. On single-place aircraft, a single on-off type of control which will provide repeated-cycle operation at a rate found optimum for the particular installation shall be provided. On multiplace aircraft, a control time will be provided which will permit a selection of the following two operating cycles:

(a) Repeat-cycle operation with no time delay or 1 minute cycle frequency for heavy or severe icing.

(b) Repeat-cycle operation with a 3-minute dwell time between cycles or a 3-minute cycle frequency for light to moderate icing.

3.12 **Performance.** De-icing boot systems when installed in the aircraft shall operate satisfactorily with the aircraft at rest, taxiing, taking off, landing, or in flight, under the following conditions:

(a) When installed in single-engine aircraft with the engine operating at 50 percent cruise manifold pressure, and with the engine operating at maximum endurance rpm for single-engine turbine-powered aircraft.

(b) When installed in twin-engine aircraft with first one and then the other of the engines operating.

(c) When installed in aircraft equipped with four or more engines with any of the two engines operating.

(d) The operation of the boots, when pulsating, shall not appreciably affect the stall characteristics or handling characteristics of the aircraft during takeoffs, landings, cruise, or high-speed conditions.

3.13 **Identification of product.** Equipment, assemblies, and parts shall be marked for identification in accordance with Standard MIL–STD–130. The following special characteristics shall be added:

Model airplane and Part No.
Date of manufacture.

3.14 **Reports.**

3.14.1 *Drawing requirements.* Drawing and engineering data covering the installation of de-icing boots, including weights and related equipment, shall conform to the requirements of Specification MIL–D–5028 and shall be furnished to the procuring activity.

3.14.2 *Product specifications.* Manufacturer shall prepare equipment specifications for all items of equipment used in the system which are not covered by detail specifications of the procuring activity, which are either designed and manufactured specifically for use in this system, or which require additional testing to prove their adequacy for use in the system. Test reports shall be submitted on all items of equipment used in the system to demonstrate compliance with the test requirements of this specification.

3.14.3 *Flight test report.* At the completion of the flight testing of the system, a report shall be submitted to substantiate the requirement specified in 3.12, when tested in accordance with 4.5.

3.15 **Workmanship.** The de-icing boot system, including all parts and accessories, shall be fabricated and finished in a thoroughly

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workmanlike manner. Particular attention shall be given to freedom from blemishes and defects, accuracy of dimensions, and marking of parts and assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests. The inspection and testing of de-icing systems shall be classified as follows:

(a) Preproduction tests (see 4.2).

(b) Acceptance tests (see 4.3).

4.2 Preproduction tests.

4.2.1 *Sampling instructions.* The preproduction samples shall consist of a representative item of each item of equipment used in the system, and shall be tested to indicate suitability of the design for the use intended in a place and manner designated in the contract, purchase order, or invitations for bids.

4.2.2 *Tests.* The preproduction tests shall be in accordance with the manufacturer's approved equipment specifications and shall include life and performance tests, plus the following Environmental tests in accordance with Specification MIL–E–5272:

(a) High temperature, Procedure I.

(b) Low temperature, Procedure I.

(c) Humidity, Procedure I.

(d) Altitude.

(e) Salt spray, Procedure I.

(f) Vibration, Procedure I (not applicable to boots).

(g) Fungus resistance, Procedure I (applicable only if any organic material is used).

(h) Sunshine, Procedure I (applicable only to boots).

4.2.2.1 *Demonstration tests.* Preproduction tests shall be performed to demonstrate satisfactorily ice removal under conditions specified in paragraph 3.7.

4.3 Acceptance tests. Each de-icing boot system installation shall be inspected and tested as specified in 4.4. In addition, each installation shall be visually inspected to insure fabrication and assembly in accordance with approved engineering design for the system and the requirements of this specification.

4.4 Ground tests. Ground tests of the system shall be conducted with the airplane at rest, using a pressure of at least 17 psi and a suction of at least 4 in. Hg to demonstrate general security and safety of the system for Flight test as specified in 4.5, and control system operation as specified in 3.11. The systems shall also be checked for leaks by inflating to 17 psi. The pressure shall not decrease by more than 5 psi in 1 minute.

4.5 Flight tests. Tests shall be conducted as part of the flight test program of the aircraft on which the boots are installed, to evaluate the effect of the boot system operation on performance and handling characteristics of the aircraft. The handling characteristics of the aircraft at various gross weights (including maximum gross weight) during takeoff, landings, cruise, and at high-speed conditions shall be demonstrated. Operation of the boots shall not cause appreciable adverse effect on the performance of the aircraft. The stall speed of the aircraft shall be substantially the same with the boots pulsating as with the boots not pulsating. The de-icing boot system shall be observed during flight in regard to the rate of cycling and uniformity of tube inflation under the following conditions:

(a) When installed in single-engine aircraft with engine operating at 50 percent cruise manifold pressure, and with the engine operating at maximum endurance rpm for single-engine turbine-powered aircraft.

(b) When installed in twin-engine aircraft with first one and then the other engine operating.

(c) When installed in aircraft equipped with four or more engines with any two engines operating.

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4.6 Rejection and retest. Material and parts which do not conform to the requirements of this specification shall be rejected. At the discretion of the manufacturer, material that has been rejected may be reworked to correct the defects and resubmitted for acceptance. Full particulars concerning previous rejection and action taken to correct the defects shall accompany all such material.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing shall be in accordance with Specification MIL–P–7936. Level A preservation for detail parts of the system shall be as specified in the detail specification. Preservation for de-icing boots shall be in accordance with MIL–P–116, method III.

5.2 Packaging.

5.2.1 *Unit boot packaging.* Each de-icing boot shall be individually packaged in containers conforming to Specification PPP–B–636 in such manner and dimensions to prevent sticking or permanent deformation of material under storage conditions.

5.3 Packing.

5.3.1 *Immediate use (for domestic shipment with immediate use at initial destination).* Packages which require overpacking for acceptance by the carrier shall be packed in commercial exterior shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet the Consolidated Freight Classification Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3.2 *Domestic shipment.* The components shall be packed in domestic-type exterior shipping containers conforming to Specification PPP–B–591, PPP–B–621, PPP–B–585, PPP–B–636, PPP–B–601 or MIL–B–10377. Exterior shipping containers shall be of minimum cube and tare consistent with the protection required. Insofar as is practicable, exterior shipping containers shall be of uniform shape and size and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Strapping and closure shall be in accordance with the appendix of the applicable container specification. When fiberboard containers are used, the fiberboard shall conform to the special requirements table of Specification PPP–B–636, as applicable.

5.3.3 *Overseas shipment.* The components shall be packed in overseas-type shipping containers conforming to Specification PPP–B–601, PPP–B–591, PPP–B–621, MIL–C–104, or MIL–B–138. Insofar as is practicable, exterior shipping containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Strapping and closures shall be in accordance with the appendix of the applicable container specification. Containers shall be provided with a case liner conforming to Specification MIL–L–10547 and sealed in accordance with the appendix thereto.

5.4 Physical protection. Cushioning, blocking, bracing, and bolting, as required, shall be in accordance with JAN–P–100, except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived. When preservation, packaging, and packing are for immediate use or when the drop tests conforming to Specification MIL–P–116 are applicable, the drop tests of JAN–P–100 shall be waived.

5.5 Marking of shipments. Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129, and as specified in the detail specification.

6. NOTES

6.1 Intended use. The de-icing equipment covered under this specification is intended

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for use on aircraft to reduce flight hazards under cold weather conditions.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type of aircraft. (see 3.12).

(c) Drawing numbers or part number of the component. (see 3.2).

(d) Whether domestic or overseas packing is required. (see 5.3).

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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